INCH-POUND
MIL-PRF-49471/12A(CR)
30 November 2000
Superseding
MIL-PRF-49471/12(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X093/U

This specification is approved for use within Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

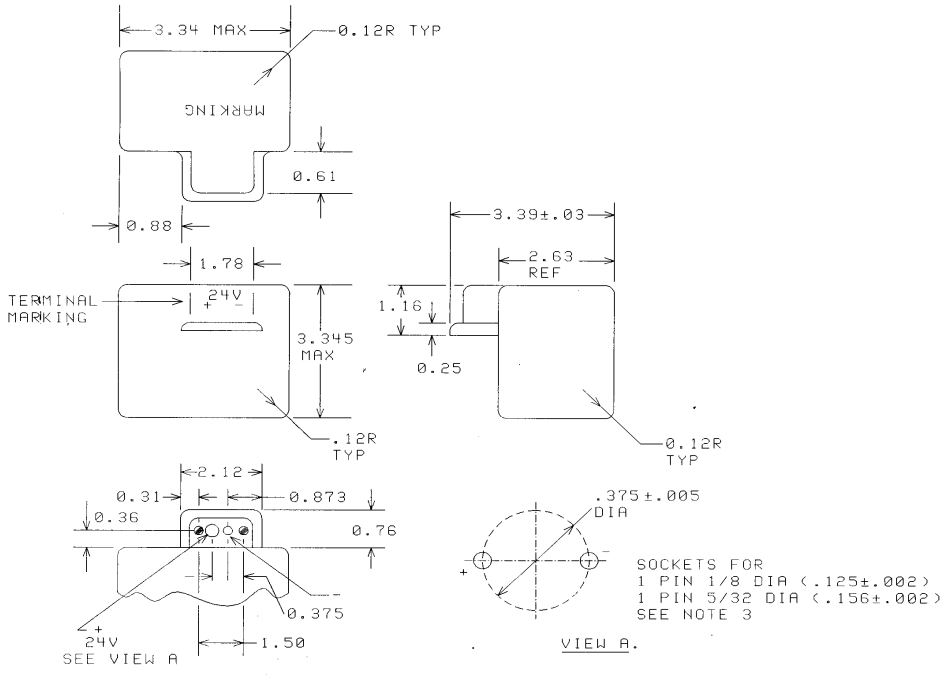

Figure 1. Battery dimensions and terminal markings.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.

NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±0.015 for three-place decimals and ±0.01 for two-place decimals.

3. Insertion Force: 12 pounds MAX. Withdrawal Force: 3 pounds MIN.

4. Socket requirements to be certified in the First Article Report.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 29.7 volts.

Cut-off voltage (for capacity): 18.00 volts.

Terminals: Flat plate type, fiberglass female socket connector with two push in terminals per Figure 1, with no obstruction of any holes.

Weight (maximum): 2.15 pounds (975 grams).

Battery short circuit test: Applicable.

Charge protection: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

State of Charge Indicator: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 18.00 Volts
I	3.4
L	1.7
H	6.8
HT	5.5
LT	1.6
IT	3.1
LR1	49.0

Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below the 18.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests, initial voltage below 18.00 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are described in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 1.0 ampere at 70 ± 5^{o}F to 2.0 volts. Each discharged cell shall then be connected in series to eight fresh cells. Each string shall be discharged at 1.0 ampere at 70 ± 5^{o}F for 3.4 hours.

Batteries: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 1.0 ampere constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests, the batteries shall be discharged through 1.0 ampere constant current for five minutes followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1, batteries shall be discharged at 75 milliamperes to 18.5 volts followed by 2.5 watts to 3 volts.

Forced Discharge: After discharge to zero volts on the I capacity test, batteries shall be force discharged at 1 ampere for a minimum of 10 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.

Overload test: Batteries shall be discharged at 2.0 amperes.

Abuse test, predischarge: Batteries shall be discharged at 1 ampere for 1.7 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 1.4 amperes for 1 minute followed by 0.9 amperes for 4 minutes, cycled continuously.

Closed-circuit voltage (battery and cell string): Load resistance of 18 ohms shall be used between the battery terminals. Voltage shall be 18.0 volts or greater within ten seconds.

Cell closed circuit voltage: Load resistance of 2 ohms shall be used. Voltage shall be above 2.0 volts within 5 seconds.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – CR

Preparing Activity:
Army - CR

(Project 6135-0417-12)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.